

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

September 2024 – December 2024

BU Spark!

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Scalable Chat Platform | Go, AWS (ECS, RDS, ElastiCache, ALB), WebSockets, Docker

[Month] 2026

- Architected cloud-native messaging platform on AWS ECS with auto-scaling policies, PostgreSQL RDS, and Redis ElastiCache, handling [X] concurrent WebSocket connections (load test with Artillery.js)
- Implemented horizontal scaling with Application Load Balancer and sticky sessions, deploying via GitHub Actions CI/CD with blue-green deployment strategy achieving [X]% uptime (CloudWatch metrics)
- Integrated comprehensive observability with CloudWatch dashboards, custom metrics, and structured logging, enabling sub-[X]ms P95 response time monitoring (APM traces and database query analysis)

Real-Time Analytics Pipeline | Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose

[Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

Audio Transcription Platform | Java Spring Boot, Python FastAPI, React, PostgreSQL

September 2025

- Built microservices platform with Spring Boot API, Python FastAPI service, and React frontend, implementing async audio processing with OpenAI Whisper and real-time status polling
- Designed PostgreSQL database with 4 related tables, Flyway migrations, and optimized indexing, supporting multiple audio formats with comprehensive error handling
- Deployed containerized application using Docker Compose with inter-service communication, CORS configuration, and health monitoring endpoints

TECHNICAL SKILLS

Cloud Platforms: AWS (ECS, RDS, ElastiCache, ALB, CloudWatch), Container Orchestration, Auto-scaling

DevOps & CI/CD: GitHub Actions, Docker, Docker Compose, Blue-Green Deployments, Infrastructure as Code

Monitoring & Observability: CloudWatch, APM Traces, Custom Metrics, Structured Logging, Performance Monitoring

Load Balancing & Scaling: Application Load Balancer, Horizontal Scaling, Sticky Sessions, Load Testing (Artillery.js)

Containerization: Docker, Docker Compose, Microservices Deployment, Health Monitoring, Service Communication

Infrastructure Management: NAS, User Account Management, Security Audits, Endpoint Security, System Uptime